Machine Elements – Exam Revision

Chapter 1: Introduction & Stress Analysis · Chapter 2: Static Failure Theories

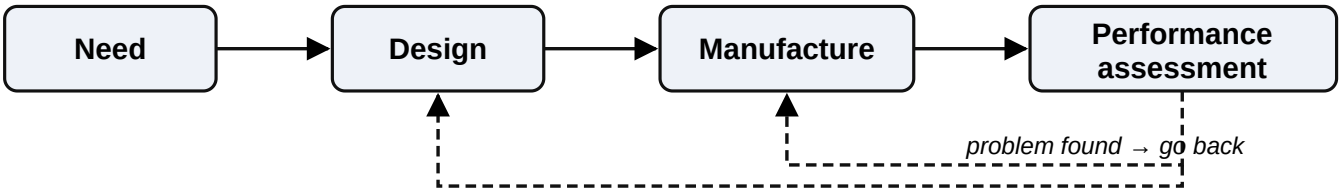
Contents

1. The design process, failure and loading
2. Stress, strain and Hooke's law
3. Stresses in a round bar (axial, shear, torsion, bending)
4. Principal stresses and 3-D Mohr circles
5. **Example 1** – Loaded cantilever cylinder (points A and B)
6. **Example 2** – Principal stresses of a 3-D stress element
7. Failure theories: max normal stress, Tresca, von Mises
8. Summary of the failure procedure
9. **Example 3** – Factor of safety at a point (Tresca & von Mises)
10. **Example 4** – Thick-walled cylinder: finding the outer radius
11. **Example 5** – Shaft diameter using von Mises
12. Formula sheet

Chapter 1 · Introduction and Stress Analysis

1. The design process

Design always starts from a **need**. After the part is designed and made, its performance is checked. If a problem appears, you go back to manufacturing or to the design.



Failure of mechanical parts can take many forms: yielding/fracture, **buckling**, chemical failure (corrosion), **creep** and **fatigue**.

Types of loading: Static loading means the load is constant with time. **Dynamic loading** means the load varies with time, which leads to fatigue.

2. Stress, strain and Hooke's law

- **Stress** = force per unit area. There are two types: **normal stress σ** (perpendicular to the surface) and **shear stress τ** (parallel to the surface).
- **Normal strain** ε = change in length / original length. **Shear strain** γ = change in angle.
- **Elastic region (Hooke's law):**

> $\sigma = E\cdot\varepsilon \qquad \tau = G\cdot\gamma \qquad$ (E = Young's modulus, G = shear modulus)

3. Stresses in a solid round bar (diameter d, outer radius c = d/2)

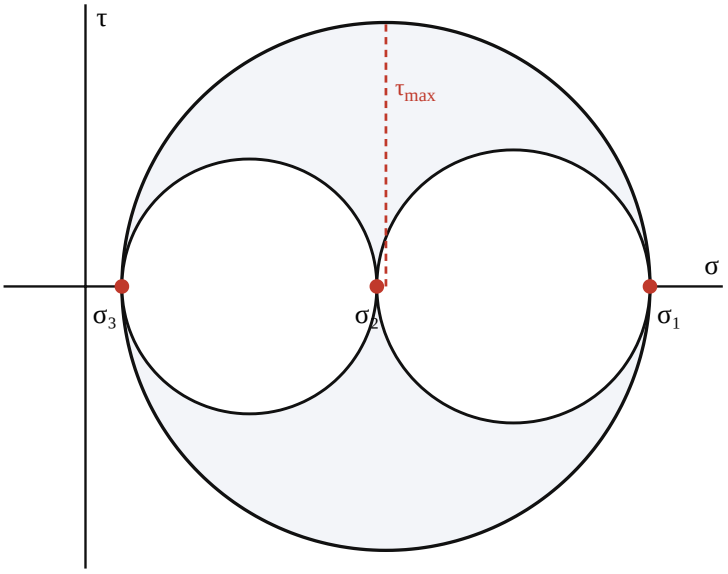
Load	Stress	Where it is maximum
Axial force F	$\sigma = F / A$	Uniform over the section
Transverse shear force V	$\tau_{max} = 4V / 3A$	At the neutral axis (zero at the top and bottom)
Torsional moment T	$\tau = T \cdot c / J = 16T / \pi d^3$	At the outer surface, all around
Bending moment M	$\sigma = M \cdot c / I = 32M / \pi d^3$	At the top and bottom (zero at the neutral axis)

> $A = \pi d^2/4 \quad I = \pi d^4/64 \quad J = \pi d^4/32 = 2I$

4. Principal stresses and 3-D Mohr circles

The stress state at a point is the stress tensor. The **principal stresses** σ_1, σ_2, σ_3 are its eigenvalues, so you find them by solving:

> $$\begin{vmatrix} \sigma_x-\sigma & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y-\sigma & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z-\sigma \end{vmatrix} = 0 \Rightarrow \sigma_1, \sigma_2, \sigma_3$$

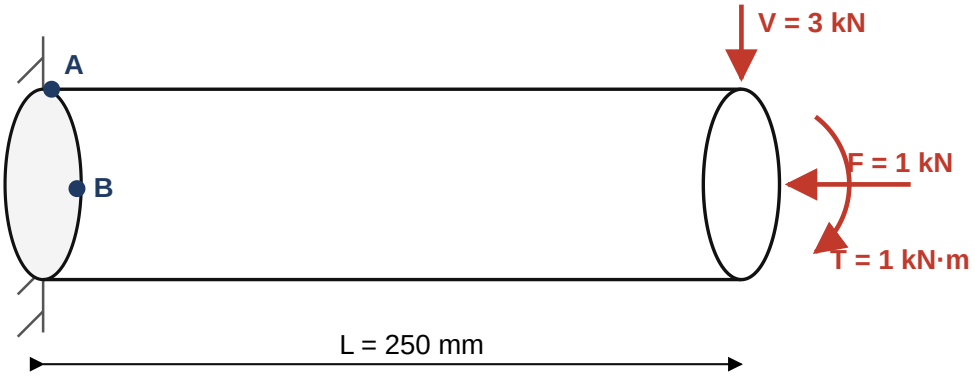


Three-dimensional Mohr circles ($\sigma_1 \geq \sigma_2 \geq \sigma_3$)

> $\tau_{max} = \max(|\sigma_1-\sigma_2|/2 , |\sigma_2-\sigma_3|/2 , |\sigma_1-\sigma_3|/2) = (\sigma_1 - \sigma_3)/2$

Example 1 – Loaded cantilever cylinder

A solid cylinder, **d = 50 mm** and **L = 250 mm**, is fixed at one end. Its free end carries an axial force **F = 1 kN**, a transverse force **V = 3 kN** and a torque **T = 1 kN·m**. Find the stresses at point **A** (top, at the wall) and point **B** (side, on the neutral axis, at the wall).



Section properties: $A = 1.96\times10^{-3}$ m², $I = 3.07\times10^{-7}$ m⁴, $J = 6.14\times10^{-7}$ m⁴, $c = 0.025$ m

- **Axial force:** $\sigma_A = \sigma_B = -F/A = -1000 / 1.96\times10^{-3} =$ **−0.509 MPa**. It is the same at every point. The minus sign is used because the notes take F as compressive.
- **Shear force:** $\tau_A = 0$ (top). At B: $\tau = 4V/3A = 4(3000) / 3(1.96\times10^{-3}) =$ **2.04 MPa**.
- **Torsion:** $\tau_A = \tau_B = Tc/J = 1000(0.025) / 6.14\times10^{-7} =$ **40.7 MPa**.
- **Bending** at the wall: $M = V\cdot L = 3$ kN × 0.25 m = 0.75 kN·m. $\sigma_A = Mc/I = 750(0.025) / 3.07\times10^{-7} =$ **61.1 MPa**, and $\sigma_B = 0$ (neutral axis).

Combined stresses:

At A: $\sigma_x = \sigma_F + \sigma_M = -0.509 + 61.1 =$ **60.6 MPa**, $\tau = \tau_T =$ **40.7 MPa**

At B: $\sigma_x =$ **−0.509 MPa**, $\tau = \tau_V + \tau_T = 2.04 + 40.7 =$ **42.8 MPa**

Check with your professor: the notes write the torsion stress as **0.04 MPa** (and so 2.08 MPa at B). That is the result if T = 1 N·m is used instead of 1000 N·m. With T = 1 kN·m, the correct torsion stress is 40.7 MPa. The steps are the same either way.

Example 2 – Principal stresses of a 3-D stress element

Given: $\sigma_x = 20$ MPa, $\sigma_y = -10$ MPa, $\sigma_z = 15$ MPa, $\tau_{yz} = 5$ MPa (all other shear stresses are 0).

$$\begin{vmatrix} 20-\sigma & 0 & 0 \\ 0 & -10-\sigma & 5 \\ 0 & 5 & 15-\sigma \end{vmatrix} = 0$$

$$(20-\sigma)\cdot[(-10-\sigma)(15-\sigma) - 25] = 0$$

$$(20-\sigma)\cdot(\sigma^2 - 5\sigma - 175) = 0$$

The first factor gives $\sigma = 20$. The quadratic gives $\sigma = (5 \pm \sqrt{725})/2 = 15.96$ or -10.96.

$\sigma_1 =$ **20 MPa**, $\sigma_2 =$ **15.96 MPa**, $\sigma_3 =$ **−10.96 MPa**

$\tau_{max} = (\sigma_1 - \sigma_3)/2 = (20 + 10.96)/2 =$ **15.48 MPa**

Chapter 2 · Static Failure Theories

7. The three failure theories (yield strength S_y, factor of safety n)

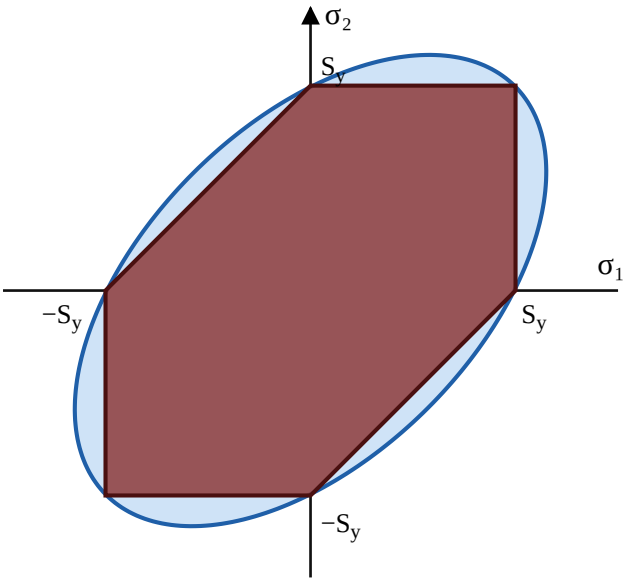
Theory	Safe condition	Factor of safety
Case 1 – Maximum normal stress	$\max(\lvert\sigma_1\rvert, \lvert\sigma_2\rvert, \lvert\sigma_3\rvert) < S_y$	$n = S_y / \max\lvert\sigma_i\rvert$
Case 2 – Maximum shear stress (Tresca)	$\max(\lvert\sigma_1-\sigma_2\rvert, \lvert\sigma_2-\sigma_3\rvert, \lvert\sigma_3-\sigma_1\rvert) < S_y$	$n = S_y / (\sigma_1 - \sigma_3)$
Case 3 – Distortion energy (von Mises)	$\sigma' = \sqrt{\{ \tfrac{1}{2}[(\sigma_1-\sigma_2)^2 + (\sigma_2-\sigma_3)^2 + (\sigma_3-\sigma_1)^2] \}} < S_y$	$n = S_y / \sigma'$

Von Mises written with stress components:

> $\sigma' = (1/\sqrt{2})\cdot\sqrt{[\ (\sigma_x-\sigma_y)^2 + (\sigma_y-\sigma_z)^2 + (\sigma_z-\sigma_x)^2 + 6(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2)\ ]}$
>
> Plane stress ($\sigma_3 = 0$): $\sigma' = \sqrt{(\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2)} < S_y$ → this is an **ellipse**
>
> Bar with σ and τ only: $\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$

Tresca vs von Mises

- **Pure shear** (only τ_{xy}): von Mises gives $\sigma' = \sqrt{3}\cdot\tau_{xy} = S_y$, so the shear yield strength is $\mathbf{S_{sy} = S_y/\sqrt{3} = 0.577\ S_y}$.
- Tresca assumes $\mathbf{S_{sy} = 0.5\ S_y}$, which is about 15 % lower than von Mises.
- So **Tresca is more conservative** (safer). Its safe region is a **hexagon** that sits inside the von Mises **ellipse**.



Plane-stress safe regions: Tresca hexagon (dark) inside the von Mises ellipse (light)

8. Summary of the failure procedure

1. Choose the **most critical point** of the part and work out all its stresses: $\sigma = (\sigma_x, \sigma_y, \sigma_z, \tau_{xy}, \tau_{yz}, \tau_{zx})$.
2. Find the **principal stresses** $\sigma_1, \sigma_2, \sigma_3$ as the eigenvalues of the stress tensor (the determinant above).
3. Pick a failure theory (Case 1, 2 or 3) and require its equivalent stress to be **smaller than S_y**.
4. Factor of safety: **$n = S_y$ / (equivalent stress)**.

Example 3 – Factor of safety at a point

Given: **$\sigma_x = 40$ MPa, $\sigma_y = 30$ MPa, $\tau_{xy} = 20$ MPa, $\tau_{yz} = 10$ MPa, $S_y = 100$ MPa** ($\sigma_z = \tau_{xz} = 0$). Find n using Tresca and von Mises.

Step 1 – Principal stresses

$$\begin{vmatrix} 40-\sigma & 20 & 0 \\ 20 & 30-\sigma & 10 \\ 0 & 10 & -\sigma \end{vmatrix} = 0$$

$$(40-\sigma)[(30-\sigma)(-\sigma) - 100] - 20[20(-\sigma)] = 0$$

$$\sigma^3 - 70\sigma^2 + 700\sigma + 4000 = 0$$

$\sigma_1 = 56.31$ MPa, $\sigma_2 = 17.71$ MPa, $\sigma_3 = -4.01$ MPa (check: the sum is $70 = \sigma_x+\sigma_y+\sigma_z$ ✓)

Step 2 – Differences: $|\sigma_1-\sigma_2| = 38.60$, $|\sigma_2-\sigma_3| = 21.72$, $|\sigma_1-\sigma_3| = 60.32$ MPa

Step 3 – Tresca

$$\text{max difference} = 60.32 \text{ MPa} \Rightarrow n = 100 / 60.32 = \mathbf{1.66}$$

Step 4 – von Mises

$$\sigma' = \sqrt{\tfrac{1}{2}[38.60^2 + 21.72^2 + 60.32^2]} = \sqrt{(\tfrac{1}{2} \times 5600)} = 52.9 \text{ MPa} \Rightarrow n = 100 / 52.9 = \mathbf{1.89}$$

Tresca gives the smaller n, so it is the more conservative of the two.

Example 4 – Thick-walled cylinder (closed ends, internal pressure P_i)

The critical point is the **inner radius**, where the stresses are:

$$\sigma_r = -P_i \quad \sigma_\theta = P_i(R_o^2 + R_i^2)/(R_o^2 - R_i^2) \quad \sigma_z = P_i R_i^2/(R_o^2 - R_i^2)$$

Here **R_o is unknown**. To see the method, take the thickness also equal to 100 mm, so $R_i = 100$ mm and $R_o = 200$ mm:

$$\sigma_r = -P_i, \quad \sigma_\theta = (40000+10000)/(40000-10000)\cdot P_i = (5/3)P_i, \quad \sigma_z = 10000/30000\cdot P_i = (1/3)P_i$$

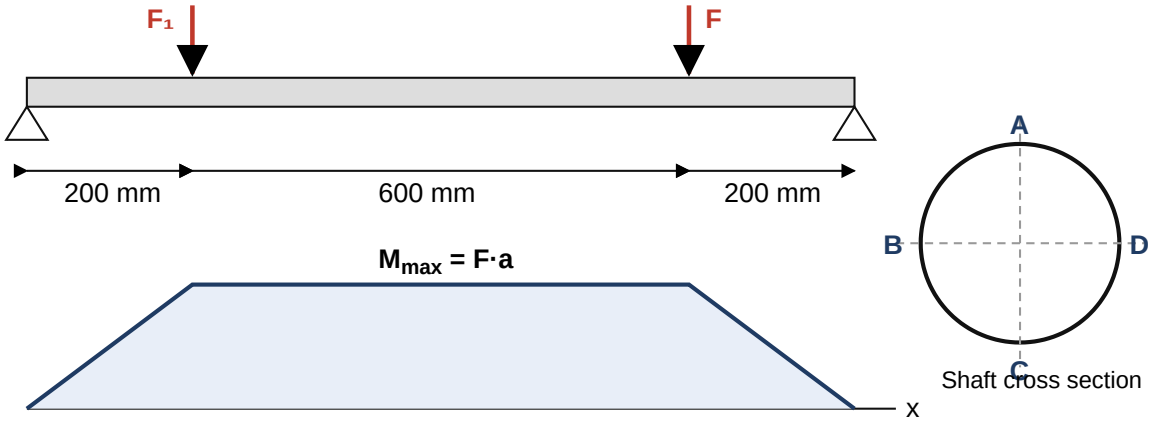
The **largest difference is between σ_θ and σ_r**, so Tresca gives:

$$n = S_y / |\sigma_\theta - \sigma_r| = 2 \quad \text{with} \quad \sigma_\theta - \sigma_r = 2P_i R_o^2/(R_o^2 - R_i^2)$$

Method: put in P_i, S_y and n, solve this equation for the unknown **R_o**, then the wall thickness is **$t = R_o - R_i$**.

Example 5 – Shaft diameter using von Mises (S_y = 400 MPa)

The shaft has bearings at both ends with spans of **200 mm – 600 mm – 200 mm**. It carries gear/pulley forces and transmits a torque T. Find the diameter d using von Mises and a factor of safety n.



Bending moment diagram – the critical section is where the force is applied

At the critical section there is a shear force F, a torque T and a bending moment M.

- **T** causes shear stress at all points on the surface: $\tau = Tc/J = 16T/\pi d^3$.
- **M** causes the largest normal stress at **A and C** (top and bottom): $\sigma = Mc/I = 32M/\pi d^3$.
- The **shear force F** causes shear stress at **B and D** (4F/3A). It is small, so it is **neglected**.
- So the critical points are **A and C**.

$$\sigma' = \sqrt{(\sigma^2 + 3\tau^2)} = (16/\pi d^3)\cdot\sqrt{(4M^2 + 3T^2)} = S_y/n$$

$$\mathbf{d = [\ 16\cdot n\cdot\sqrt{(4M^2 + 3T^2)} \,/\, (\pi\cdot S_y)\ ]^{1/3}}$$

Steps: (1) draw the bending moment diagram and read M_{max}; (2) get T from the power or gear forces; (3) put M, T, S_y = 400 MPa and n into the formula; (4) round d up to a standard size.

> The actual force and torque values in the notes were not readable in the photo. Use the numbers from the sheet with the formula above.

Formula sheet

Item	Formula
Hooke's law	$\sigma = E\varepsilon, \quad \tau = G\gamma$
Round bar	$A = \pi d^2/4, \quad I = \pi d^4/64, \quad J = \pi d^4/32$
Axial / bending	$\sigma = F/A, \quad \sigma = Mc/I = 32M/\pi d^3$
Torsion / transverse shear	$\tau = Tc/J = 16T/\pi d^3, \quad \tau_{max} = 4V/3A$
Principal stresses	$\det[\sigma_{ij} - \sigma\delta_{ij}] = 0$
Max shear (3-D Mohr)	$\tau_{max} = (\sigma_1 - \sigma_3)/2$
Max normal stress	$n = S_y / \max\lvert\sigma_i\rvert$
Tresca	$n = S_y / (\sigma_1 - \sigma_3), \quad S_{sy} = 0.5S_y$
von Mises	$\sigma' = \sqrt{\{\tfrac{1}{2}[(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2]\}}, \quad n = S_y/\sigma', \quad S_{sy} = 0.577S_y$
von Mises (σ and τ)	$\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$
Thick cylinder (inner radius)	$\sigma_r = -P_i, \quad \sigma_\theta = P_i(R_o^2+R_i^2)/(R_o^2-R_i^2), \quad \sigma_z = P_iR_i^2/(R_o^2-R_i^2)$
Shaft diameter (von Mises)	$d = [16n\sqrt{(4M^2+3T^2)}/(\pi S_y)]^{1/3}$